

Canvas From The Sky

Supplementary calculation document for the investor pitch deck

150 m peak operating height · 25-guest commercial cap · 20-minute per-batch night cycle · baseline ARPU retained

Audience	Company executive review — read alongside PitchDeck v2 (index.html)
Date	15 May 2026
Status	Aligned with the revised PitchDeck v2 HTML
Sources	pitch-blueprint.md (baseline) · financial-model-revision-150m.md (working draft)

1. Purpose of this memo

The investor pitch deck (PitchDeck v2, index.html) communicates the revised Canvas From The Sky financial model in summary form. This memo is the calculation backing it. Each headline figure in the deck can be traced here to the operating assumption it rests on, and each operating assumption is shown against the previous pitch-blueprint.md baseline so executives can see what changed and why.

The revision moves three operating assumptions versus the pitch-blueprint.md baseline:

- Maximum balloon height: 300 m → 150 m (true new peak — the attraction is no longer marketed as 300 m capable).
- Commercial basket loading: 30 guests → 25 guests (the physical gondola remains 30-passenger).
- Night cycle: a single 41-minute cycle → a 20-minute per-batch cadence, with an on-platform photo dwell after the gondola lands and the LED entry hallway closed during the main show.

2. At a glance — baseline vs. revised

A single-table view of every assumption and outcome that moved between the pitch-blueprint.md baseline and the revised model in PitchDeck v2. Sections §3 onward show the calculations behind each row.

Dimension	Baseline (blueprint)	Revised (this memo)	Change
Operating assumptions			
Peak balloon altitude	300 m	150 m	True new peak
Gondola commercial capacity	30 guests / batch	25 guests / batch	– 5 guests
Operating days (day flights)	300 / year	250 / year	– 50 days
Operating nights (LED show)	300 / year	250 / year	– 50 nights
Day cycle	30 min · 6 rides/day	30 min · 6 rides/day	Unchanged
Night cycle	41 min · 8 rides/night	20 min · 16 batches/night	+ 8 batches/night
Day blended ARPU	AED 170	AED 170	Unchanged
Night blended ARPU	AED 361	AED 361	Unchanged
Base case load factor	70 %	58 %	– 12 pts
Capacity outputs			
Day seat capacity	180 / day	150 / day	– 16.7 %
Night seat capacity	240 / night	400 / night	+ 66.7 %
Financial outputs (Base Case)			
CAPEX (single-spec aluminum)	AED 25.9 M	AED 25.9 M	Unchanged
Annual revenue	AED 24.7 M	AED 22.2 M	– AED 2.5 M
EBITDA	AED 14.9 M	AED 12.4 M	– AED 2.5 M
EBITDA margin	60 %	56 %	– 4 pts
Simple payback	~1.6 yrs	~2.1 yrs	+ 0.5 yrs
10-year IRR	~58 %	~35.0 %	– 23 pts
10-year NPV at 8 %	AED 85 M	AED 51.4 M	– AED 33.6 M
10-year MOIC	~5.5×	~4.7×	– 0.8×

Read in one line: the revised model trades a meaningful share of the baseline upside for a more conservative operating envelope. The night cycle doing 16 batches at 25 guests is the load-bearing change — without it, the 250-day operating year and the 25-guest cap would have pushed returns well below the figures shown.

3. Baseline reference — pitch-blueprint.md

The previous investor model documented in pitch-blueprint.md (Part 1, Part 7, Part 8) used the following operating assumptions and outputs. These are the figures the revision is measured against.

Operating assumptions (baseline)

Parameter	Baseline value	Source in blueprint
Maximum balloon height	300 m	Part 1 · Appendix A
Gondola commercial capacity	30 guests	Part 1 · Appendix A
Operating days (day flights)	300 / year	Part 7
Operating nights (LED show)	300 / year	Part 7
Day flight cycle	30 min · 6 rides/day	Part 1
Night LED show cycle	41 min · 8 rides/night	Part 1
Day blended ARPU	AED 170	Part 7
Night blended ARPU	AED 361	Part 7
Base case load factor	70 %	Part 7
All-Access pass usage factor	1.4 rides / unique guest	Part 7

Baseline financial outputs

Metric	Baseline
Base annual revenue	AED 24.7 M
Strong-case revenue	AED 39.7 M
Base annual OpEx	AED 9.8 M
Base EBITDA / margin	AED 14.9 M · 60 %
Base simple payback	~1.6 years
10-year IRR	~58 %
10-year NPV at 8 %	AED 85 M
10-year MOIC	~5.5×

CAPEX in the baseline is AED 25.9 M for the High Tier (aluminum tile) and AED 25.3 M for the Low Tier (GFRC tile). This memo uses the AED 25.9 M figure to preserve a single-spec comparison.

4. Revised operating assumptions

Parameter	Revised value	Rationale
Maximum balloon height	150 m	New true peak. Lower operating altitude under typical Saadiyat wind/safety conditions. The product is no longer marketed as 300 m-capable.
Gondola commercial capacity	25 guests	More conservative than the 30-pax physical capacity. Improves guest comfort and gives an operating margin for staff and accessibility cases.
Operating days (day flights)	250 / year	Higher weather-cancellation and operational-downtime allowance than the baseline 300 days. ~32 % cancellation buffer.
Operating nights (LED show)	250 / year	Same buffer as day operations.
Day flight structure	6 rides / day	Unchanged from baseline. Daytime experience is the museum-and-skyline panorama; no LED show, so cycle compression is not needed.
Night cycle	20 minutes per batch	Faster cycle becomes feasible because ascent and descent are shorter at 150 m. See section 5 for the per-batch timeline.
Night batches per night	16 / night	3 batches per hour × ~5.5-hour night window = 16.5, rounded down for buffer.
All-Access pass usage factor	1.4 rides / unique guest	Unchanged from baseline. The 20-minute cycle actually improves the All-Access proposition (multi-theme viewing is easier in one night).
Base case load factor	58 %	Below the baseline 70 % Dubai Balloon benchmark. Acknowledges a slower ramp-up and a more conservative steady-state target.
Day blended ARPU	AED 170	Unchanged. See section 7 for the rationale for retaining ARPU at 150 m.
Night blended ARPU	AED 361	Unchanged. See section 7.
CAPEX (single-spec aluminum)	AED 25.9 M	High-tier aluminum tile assumption from the blueprint.

5. Night cycle design at 150 m

The 20-minute per-batch cadence is the operational hinge of the revised model. It works because a 150 m peak shortens ascent and descent versus the 300 m baseline, freeing time for a compressed show window. Two operating rules protect the experience:

- The LED entry hallway is closed in both directions during the main show. Guests cannot exit through it once the gondola has descended to the 25 m show altitude, so no one casts shadows across the canvas that the gondola is viewing from above.
- After the gondola lands, guests stay on the boarding platform for a short photo dwell. This gives them a soft, memorable exit beat — and it keeps them off the LED hallway until the next batch has cleared it.

Per-batch timeline (T = minutes from batch start)

Phase	Time	Duration	LED hallway	What happens
Boarding from LED hallway	0:00 – 2:00	~2.0 min	Open (entry)	Guests walk in across the lit hallway, board the gondola, safety brief.
Ascent to 150 m	2:00 – 5:00	~3.0 min	Open (ambient)	50 m/min ascent. LED field dims under a rising animation.
Peak hold at 150 m	5:00 – 7:30	~2.5 min	Open (ambient)	Three museums + Gulf coastline + Abu Dhabi skyline in view.
Descent 150 m → 25 m	7:30 – 10:30	~3.0 min	Open (anticipation)	40 m/min descent. LED canvas grows in resolution below the gondola.
Main show at 25 m	10:30 – 18:00	~7.5 min	CLOSED both ways	Gondola holds at 25 m. Full 55,575-node themed show. No hallway movement.
Final descent to ground	18:00 – 18:40	~0.7 min	Closed	Show closes as the gondola lands at the platform.
Photo dwell on platform	18:40 – 20:00	~1.3 min	Open (exit / next entry)	Guests stay on the platform for photos. Hallway reopens for the next batch entry; previous batch exits during the next batch's boarding window.

Why 20 minutes (and not less)

- Ascent + descent at the documented balloon speeds (50 m/min up, 40 m/min down) consumes ~6 minutes round-trip at 150 m. Below that we are not selling a flight — we are selling a lift.
- A 7.5-minute main show is the floor for a recognisable themed narrative. Shorter than that and the All-Access pass (three shows in a night) loses meaning.
- Photo dwell + boarding turnover need ~3 minutes combined for a safely-managed gondola hand-off at 25 guests/batch.

Why the hallway must close during the show

The main show runs with the gondola held at 25 m, looking straight down at the 60 m LED ring. Any pedestrian movement on the LED hallway — even a single person — produces a moving dark shape across the canvas that the gondola sees clearly from 25 m above. Closing the hallway during the show is non-negotiable for show quality; the on-platform photo dwell is what makes that closure operationally invisible.

6. Capacity impact

The 25-guest cap reduces per-batch loading, but the 20-minute cycle more than doubles nightly batch count. The net effect is a substantial gain in night-side seat capacity.

Capacity line	Baseline	Revised	Change
Day seat capacity	$6 \times 30 = 180$ / day	$6 \times 25 = 150$ / day	– 16.7 %
Night seat capacity	$8 \times 30 = 240$ / night	$16 \times 25 = 400$ / night	+ 66.7 %
Annual seat days (250 vs 300)	300 / year	250 / year	– 16.7 %

Net: night-side annual paid seats increase relative to baseline once the 250-day operating year is applied; day-side annual paid seats decline. Because night ARPU is more than 2× day ARPU, the revenue mix tilts further toward night — making the night-cycle assumption the dominant lever in the model (see section 11 sensitivity).

7. Pricing assumption — ARPU held at baseline

Day blended ARPU is held at AED 170 and night blended ARPU at AED 361, i.e. unchanged from the pitch-blueprint.md baseline. The defensible rationale at a 150 m peak is:

- Night value is driven by the LED canvas show, the three museum-tied themes, and the Saadiyat night-destination ecosystem — not by raw altitude. The show is paid for; the altitude is the delivery vehicle.
- 150 m sits within the global peer band for tethered/observation balloons: London Eye 135 m, Singapore Flyer 165 m. The premium-experience price point is defensible at this band.
- The All-Access pass (AED 695) is grounded in seeing all three themed shows in one evening. Altitude is irrelevant to the upsell.
- Day flight pricing (AED 195 adult) is justified by the unobstructed museum-and-skyline panorama at 150 m, not by altitude alone.

ARPU line	Baseline 300 m	Revised 150 m	Change
Day blended ARPU	AED 170	AED 170	0 %
Night blended ARPU	AED 361	AED 361	0 %

A moderate ARPU haircut (AED 140 day / AED 320 night) is carried only as a downside sensitivity in section 9 for prudence — it is not the headline model.

8. Revenue build — formulas and worked numbers

Ticket revenue formulas

Day ticket revenue =

$$6 \text{ rides/day} \times 25 \text{ guests} \times \text{load} \times 250 \text{ operating days} \times \text{AED } 170 \text{ ARPU}$$

Night ticket revenue =

$$16 \text{ batches/night} \times 25 \text{ guests} \times \text{load} \times 250 \text{ operating nights} \times \text{AED } 361 \text{ ARPU} \div 1.4 \text{ rides per unique guest}$$

Worked example — Base Case (58 % load)

- Day: $6 \times 25 \times 0.58 \times 250 \times 170 = \text{AED } 3.70 \text{ M}$
- Night: $(16 \times 25 \times 0.58 \times 250) \div 1.4 \times 361 = (58,000 \div 1.4) \times 361 = 41,429 \text{ unique guests} \times \text{AED } 361 = \text{AED } 14.96 \text{ M} \approx \text{AED } 15.0 \text{ M}$

Load factors used per scenario

Scenario	Load	Positioning
Conservative	32 %	Ramp-up / weather-constrained resilience case.
Base Case	58 %	Practical steady operating case below the baseline 70 % benchmark.
Strong	73 %	Demand-strong case, still below the baseline full-upside forecast.

Revenue by scenario (annual)

Revenue stream	Conservative	Base Case	Strong
Day ticket revenue	AED 2.0 M	AED 3.7 M	AED 4.7 M
Night ticket revenue	AED 8.3 M	AED 15.0 M	AED 18.8 M
F&B / retail	AED 0.3 M	AED 0.7 M	AED 1.1 M
Sponsorship — content + branding	AED 0.7 M	AED 1.6 M	AED 2.2 M
Museum partnership + cross-sell	AED 0.1 M	AED 0.4 M	AED 0.5 M
Private gondola + VIP events	AED 0.4 M	AED 0.8 M	AED 1.0 M
Total annual revenue	AED 11.8 M	AED 22.2 M	AED 28.3 M

Non-ticket revenue lines are trimmed versus the baseline blueprint, reflecting a more conservative sponsor-and-event ramp.

9. EBITDA and payback

OpEx is held close to the baseline structure documented in pitch-blueprint.md Part 4 — AED 9.5 M / 9.8 M / 10.1 M across Conservative / Base / Strong, reflecting variable staff, F&B and event-operations loading.

Scenario	Revenue	OpEx	EBITDA	Margin	Simple payback
Conservative	AED 11.8 M	AED 9.5 M	AED 2.3 M	19 %	~11.3 yrs
Base Case	AED 22.2 M	AED 9.8 M	AED 12.4 M	56 %	~2.1 yrs
Strong	AED 28.3 M	AED 10.1 M	AED 18.2 M	64 %	~1.4 yrs

Simple payback is computed against the AED 25.9 M aluminum-tier CAPEX. Below the conservative case the model loses payback discipline very quickly — see section 11 for the night-window sensitivity.

10. 10-year return estimate (Base Case)

The 10-year cash-flow ramp uses these assumptions:

- Year 1 lands at the Conservative ramp-up EBITDA.
- Year 2 reaches 85 % of Base EBITDA.
- Year 3 reaches Base EBITDA.
- Years 4 – 10 grow at 4 % per year.
- AED 2.5 M balloon envelope replacement reserves are booked in Years 5 and 10.

Metric	Revised Base Case
10-year IRR	~35.0 %
10-year NPV at 8 %	~AED 51.4 M
10-year MOIC	~4.7×

11. Downside sensitivity — reduced 150 m ARPU

If the executive review wants a more conservative pricing posture, the model below applies a moderate ARPU haircut. This is carried for prudence; it is not the recommended headline assumption.

ARPU line	Main case	Downside sensitivity		
Day blended ARPU	AED 170	AED 140		
Night blended ARPU	AED 361	AED 320		
Scenario	Revenue	EBITDA	Margin	Simple payback
Conservative	AED 10.5 M	AED 1.0 M	9 %	~26.0 yrs
Base Case	AED 19.8 M	AED 10.0 M	51 %	~2.6 yrs
Strong	AED 25.3 M	AED 15.2 M	60 %	~1.7 yrs

Base-case 10-year returns under this downside pricing sensitivity are approximately 27.7 % IRR, AED 35.3 M NPV at 8 %, and 3.7× MOIC.

12. Sensitivity — night operating window

The 20-minute cycle is the central operational lever of the revised model. If it is approved but the nightly operating window is shorter than the assumed 5.5 hours, the headline returns move materially.

Night batches / night	Base revenue	Base EBITDA	Simple payback	10-year IRR
9 batches	AED 15.6 M	AED 5.8 M	~4.5 yrs	~12.4 %
12 batches	AED 18.4 M	AED 8.6 M	~3.0 yrs	~23.1 %
15 batches	AED 21.2 M	AED 11.4 M	~2.3 yrs	~32.2 %
16 batches	AED 22.2 M	AED 12.4 M	~2.1 yrs	~35.0 %

The model assumes 16 batches/night. If operational reality settles at 12 – 15 batches, returns remain credible but compress sharply; below 9 batches the night business is no longer the load-bearing column.

13. Key takeaways for the executive read

- 150 m is the true new peak. The deck and this memo are consistent on this point — the attraction is no longer marketed as 300 m capable.
- The 25-guest cap improves guest comfort and gives operating margin. It reduces seat capacity per batch by 16.7 %.
- The 20-minute night-batch cadence is what makes the revised model work. It more than doubles nightly batch count and offsets both the 25-guest cap and the lower 250-day operating year.
- The LED hallway closes during the main show. Guests stay on the boarding platform for photos afterwards. This is the operating rule that protects show quality without losing nightly capacity.
- ARPU is held at the baseline AED 170 day / AED 361 night because value is driven by the LED show + museum themes + Saadiyat ecosystem, not by raw altitude. A moderate ARPU haircut is carried as a downside sensitivity only.
- The revised base case still supports a credible investment story: AED 22.2 M revenue, AED 12.4 M EBITDA, ~2.1-year payback, ~35 % 10-year IRR, AED 51.4 M NPV at 8 %, 4.7× MOIC.

Appendix A — Cross-reference to PitchDeck v2 (index.html)

This appendix lists every headline figure in the revised deck that this memo is the source of. Where the deck currently shows a value inherited from an earlier revision and not yet updated to the figures below, that gap is flagged for follow-up.

Deck location	Headline figure (this memo)	Notes
Slide 1 hero — altitude stat	150 m	Replaces previous 300 m stat.
Slide 3 — night ride cycle length	20-minute batch	Replaces previous ~41-minute single cycle.
Slide 3 — peak altitude step	150 m	Replaces previous 300 m peak.
Slide 3 — descent narrative	125 m at 40 m/min	Replaces previous "275 m at 40 m/min".
Slide 3 — finale narrative	Photo dwell on platform	New beat: hallway-closed-during-show is operationally invisible because guests linger on the platform.
Slide 6 — CAPEX (single-spec aluminum)	AED 25.9 M	Unchanged from blueprint.
Slide 7 — annual revenue (Base)	AED 22.2 M	Replaces interim AED 19.6 M figure if still present.
Slide 7 — EBITDA (Base)	AED 12.4 M	Replaces interim AED 9.8 M figure if still present.
Slide 7 — 10-year IRR (Base)	~35 %	Replaces interim ~27 % figure if still present.
Slide 7 — 10-year NPV at 8 %	AED 51.4 M	Replaces interim AED 34 M figure if still present.
Slide 7 — 10-year MOIC	4.7×	Replaces interim 3.6× figure if still present.

— end of memo —